Ridgeline Schools Weather Network: Health Report

Spring 2025 Coordinator Audit | Generated 2026-10-24 15:31 UTC | Stations: 8 | Flags: 16

1. Climate Summary Verdict: CONDITIONALLY USABLE

2. Priority Technician Recommendation: STATION S03 (Valley Library) [Score: 225.0]

Station	Channel	Period (UTC)	Action	Evidence & Defensible Treatment (All 16 Generated Flags)
S02	pressure_hpa	2025-03-21 00:00 to 2025-04-11 23:50	CORRECT	[uncalibrated sensor replacement] Uncalibrated pressure board swap created a +5.6 hPa step offset. Correct by applying -5.6 hPa.
S03	temp_c	2025-03-05 14:20 to 2025-03-05 14:20	EXCLUDE	[hardware error code spike] temp_c reported digital fault code -40.0°C (isolated sensor error spike).
S03	temp_c	2025-03-12 03:40 to 2025-03-12 03:40	EXCLUDE	[hardware error code spike] temp_c reported digital fault code 85.0°C (isolated sensor error spike).
S03	temp_c	2025-03-18 06:00 to 2025-03-21 14:00	EXCLUDE	[stuck sensor] temp_c frozen constantly at 6.5 for 80.0 hours (481 readings).
S03	temp_c	2025-04-02 22:10 to 2025-04-02 22:10	EXCLUDE	[hardware error code spike] temp_c reported digital fault code -40.0°C (isolated sensor error spike).
S03	temp_c	2025-04-08 09:50 to 2025-04-08 09:50	EXCLUDE	[hardware error code spike] temp_c reported digital fault code 85.0°C (isolated sensor error spike).
S04	temp_c	2025-03-15 00:00 to 2025-04-11 23:50	KEEP	[post repainting microclimate / albedo] Garden post repainted white; minor +0.4°C daytime mean change is within natural spring variance. Retain data.
S04	rain_mm	2025-03-25 16:10 to 2025-04-02 16:00	EXCLUDE	[clogged rain funnel] Rain gauge clogged with debris; missed ~31.2 mm regional precipitation prior to clearing.
S04	rain_mm	2025-04-02 16:10 to 2025-04-02 16:30	EXCLUDE	[artificial flush dump after gauge cleaning] Artificial 33.4 mm rain surge across 3 intervals caused by funnel flush during/after cleaning.
S05	metadata	2025-03-01 00:00 to 2025-04-11 23:00	CORRECT	[metadata coordinate error] Installed metadata lat 39.5700 disagrees with GPS position 39.7540 by 20.4 km. Update station latitude to 39.7540.
S05	rh_pct	2025-03-07 12:10 to 2025-03-11 00:00	EXCLUDE	[sensor saturation / out of bounds] Relative humidity exceeded 100% (451 readings, peak 105.0%). Sensor saturated/uncalibrated.
S06	all	2025-03-24 10:10 to 2025-03-24 16:00	EXCLUDE	[clock rollback / duplicate interval] Logger timestamp jumped backward by 5.8 hours (2025-03-24 16:00 to 2025-03-24 10:10).
S06	all	2025-03-24 11:00 to 2025-04-11 17:00	CORRECT	[clock timezone offset] Logger clock set to local time (-6.0h offset vs GPS UTC). Correct by shifting timestamps by +6h.
S07	wind_kmh	2025-03-26 00:00 to 2025-04-11 23:50	CORRECT	[anemometer calibration mismatch] Replacement anemometer output dropped by factor of 3.1x vs network (m/s vs km/h mismatch). Rescale by x3.6.
S08	all	2025-03-16 14:00 to 2025-03-18 01:30	EXCLUDE	[data gap] Transmission gap of 35.8 hours (214 missing records).
S08	all	2025-04-05 00:00 to 2025-04-05 23:50	EXCLUDE	[duplicate transmission block] Repeated block of 144 records identical to 2025-04-05 00:00 appended at end of file.